Question 1

```
import seaborn as sns
from scipy import stats

df = sns.load_dataset("titanic")

male = df[df["sex"] == "male"]["fare"].dropna()
female = df[df["sex"]== "female"]["fare"].dropna()

t_stat, p_value = stats.ttest_ind(
    male, female, equal_var=False
)

print("t_statistic:",t_stat)
print("p-value:", p_value)
```

```
t_statistic: -5.077499013458909
p-value: 5.388057409693902e-07
```

Question 2

```
from scipy.stats import chi2_contingency
import pandas as pd

table = pd.crosstab(df["class"], df["survived"])
print(table)

chi2, p_value, dof, expected = chi2_contingency(table)

print("Chi-square statistics:",chi2)
print("Degrees of Freedom:",dof)
print("p-value:",p_value)
```

```
survived    0    1
class
First      80  136
Second     97   87
Third     372  119
Chi-square statistics: 102.88898875696056
Degrees of Freedom: 2
p-value: 4.549251711298793e-23
```

Both status are associated

h0: mean fares do not differ significance across 3 passenger classes

```
from scipy.stats import f_oneway

first = df[df["class"] == "First"]["fare"].dropna()
second = df[df["class"]== "Second"]["fare"].dropna()
third = df[df["class"]== "Third"]["fare"].dropna()

F, p_value = f_oneway(first, second, third)

print("F-statistic:",F)
print("p-value:",p_value)
```

```
F-statistic: 242.34415651744814
p-value: 1.0313763209141171e-84
```

```
from statsmodels.stats.multicomp import pairwise_tukeyhsd

df=df.dropna(subset=["fare"])
tukey=pairwise_tukeyhsd(endog=df["fare"],groups=df["class"])
print(tukey)
```

```
 Multiple Comparison of Means - Tukey HSD, FWER=0.05
====================================================
group1 group2 meandiff p-adj  lower    upper   reject
----------------------------------------------------
 First Second -63.4925   0.0 -72.9165 -54.0685   True
 First  Third -70.4791   0.0 -78.1489 -62.8094   True
Second  Third  -6.9866 0.108 -15.1064   1.1331  False
----------------------------------------------------
```